# 3E 1 INTRODUCTION TO SOLID LATTICES

SPECIFICATION REFERENCE 3.12

## LEARNING OBJECTIVES

- Know that giant lattices are present in: solid metals (giant metallic lattices); ionic solids (giant ionic lattices); and covalently bonded solids, such as diamond, graphite and silicon(IV) oxide (giant covalent lattices).
- Know the different structures formed by carbon atoms, including graphite, diamond and graphene.
- Know that the structure of solid iodine is discrete (simple) molecular.

The four types of solid structures we shall deal with in this topic are:

- giant metallic lattices
- giant ionic lattices
- giant covalent lattices
- discrete (simple) molecular lattices.

**DID YOU KNOW?**

**There is one other important type of solid structure described as 'polymeric'. These solids are composed of macromolecules such as natural polymers and synthetic (made by humans) polymers. We will look at these in Topic 5B.**

## METALLIC LATTICES

Metallic lattices are composed of a regular arrangement of positive metal ions (cations) surrounded by delocalised electrons.

Substances that have a giant metallic lattice typically have the following properties:

- high melting and boiling temperatures
- good electrical conductivity
- good thermal conductivity
- malleability
- ductility.

We explained these properties in **Topic 3D**.

## GIANT IONIC LATTICES

Giant ionic lattices are composed of a regular arrangement of positive and negative ions.

Substances that consist of giant ionic lattices typically have the following properties:

- fairly high melting temperatures
- brittleness
- poor electrical conductivity when solid but good when molten
- often soluble in water.

We explained these properties in **Topic 3D**.

## GIANT COVALENT LATTICES

Giant covalent lattices are sometimes called network covalent lattices. They consist of a giant network of atoms linked to each other by covalent bonds.

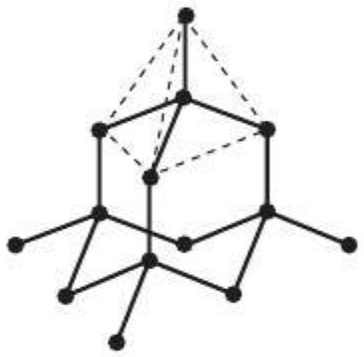
Four of the most common giant covalent substances are:

- diamond
- graphite
- graphene
- silicon(IV) oxide (not discussed below).

## DIAMOND

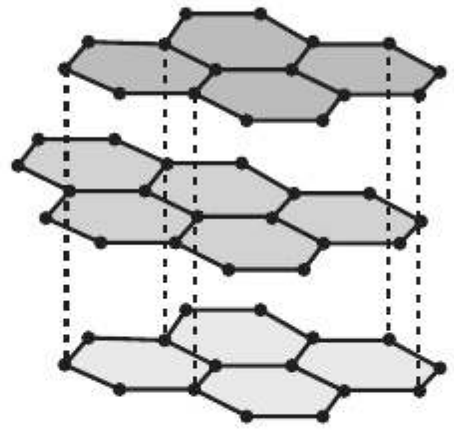
In diamond, each carbon atom forms four sigma (δ) bonds to four other carbon atoms, in a giant three-dimensional tetrahedral arrangement (**fig A**). All bond angles are 109.5°.

Diamond is extremely hard because of the very strong C—C bonding throughout the structure. It also has a very high melting temperature because a great number of strong C—C bonds have to be broken in order to melt it. This requires a large amount of heat energy.



● = a carbon atom

▲ **fig A** Structure of diamond.



● = a carbon atom

▲ **fig B** Structure of graphite.

## GRAPHITE

Graphite has a layered structure as shown in **fig B**.

Each carbon atom is bonded to three others by sigma bonds, forming interlocking hexagonal rings. The fourth electron on each carbon atom is in a p-orbital. The carbon atoms are close enough for the p-orbitals to overlap with one another to produce a cloud of delocalised electrons, both above and below the plane of the rings. We will compare this with the structure of benzene in **Topic 18 (Book 2: IAL)**.

### ADDITIONAL READING

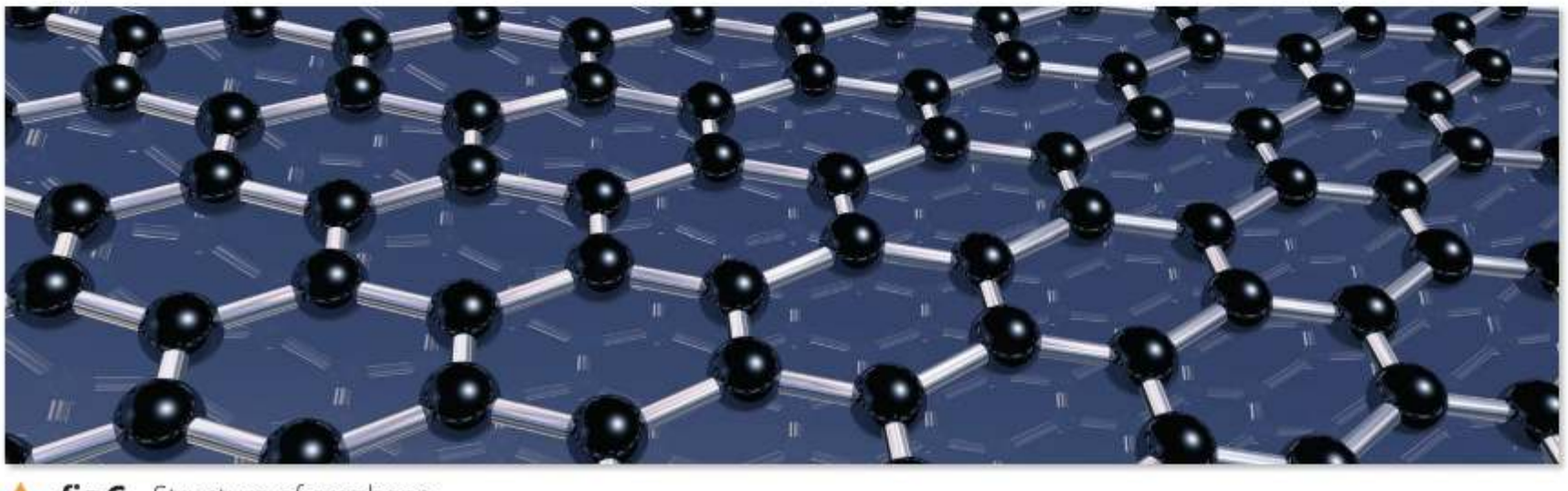
Graphite can be used as a solid lubricant since the layers slide easily over one another. Although there are only relatively weak London forces of attraction between the layers (see **Topic 7**), this does not account for its lubricating properties. Graphite is five times less able to act as a lubricant at high altitude, and eight times less in a vacuum. Its lubricating properties are as a result of adsorbed gases on the surface of the carbon atoms. Graphite's inability to act as an effective lubricant in a vacuum is the reason why it is not used as a lubricant in spacecraft. Either molybdenum disulfide ($MoS_2$) or hexagonal boron nitride (BN) can be used instead.

Graphite is a fairly good conductor of electricity. The delocalised electrons between the layers are free to flow under the influence of an applied potential difference. An interesting feature of graphite is that it can only conduct electricity parallel to its layers. The delocalised electrons are not free to move from one layer to the next. Compare this with a metal, which is able to conduct electricity in all directions throughout the structure.

Graphite has a high melting temperature for the same reason as diamond.

## GRAPHENE

Graphene is pure carbon in the form of a very thin sheet, one atom thick (**fig C**). The carbon atoms are bonded in exactly the same way as in graphite and it can, therefore, be described as a one-atom thick layer of graphite.



▲ **fig C** Structure of graphene.

### DID YOU KNOW?

You may have heard diamond referred to as 'ice'. Diamond first acquired this nickname because of its properties rather than its appearance. Owing to the wave nature of electrons, the electron wave is able to travel more smoothly through a perfect crystal without bouncing, in much the same way as light travels through a clear crystal. In addition, thermal conduction is improved by the network of strong covalent bonds. The use of a heat probe is one of the major tests in determining if a diamond is authentic. This is because real diamonds are able to conduct heat and are actually one of the best conductors of heat. Even as a diamond heats up, it is cool to the touch, just like ice. This is why the points of diamond cutting tools do not become overheated.

**DID YOU KNOW?**

Graphene is the thinnest material on Earth

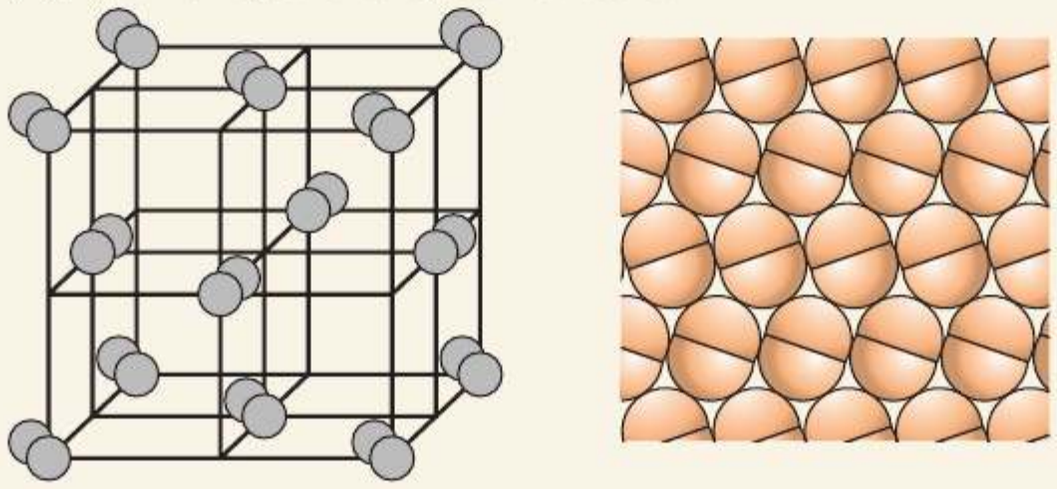
- It is one million times thinner than a human hair.
- It is 200 times stronger than steel.
- It is an excellent thermal conductor, better even than diamond.
- It absorbs light.
- It can be considered the basic unit of most other forms of carbon.
- Graphite consists of layers of graphene on top of one another joined by London forces.
- A sheet of graphene can be rolled into a ball to produce fullerene molecules.
- A sheet of graphene can be rolled into a cylinder to produce a carbon nanotube.
- Graphene can self-repair holes in its sheets when exposed to molecules containing carbon, such as hydrocarbons. When bombarded with pure carbon atoms, the atoms perfectly align into hexagons, completely filling the holes.

## ADDITIONAL READING

### Molecular lattices

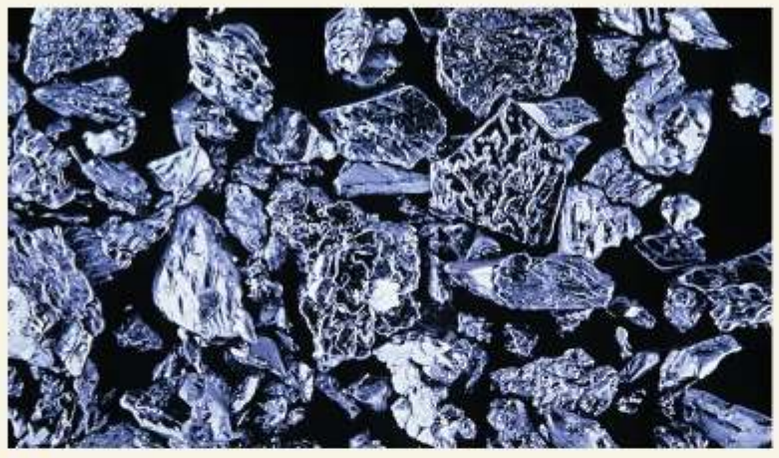
A common solid molecular lattice is iodine.

Iodine exists as diatomic molecules, $I_2$. In solid iodine these molecules are arranged in a regular pattern (**fig D**), which explains its crystalline nature.



▲ **fig D** Structure of iodine.

The diagram on the left of **fig D** shows the arrangement of the iodine molecules. The structure is described as 'face-centred cubic'. In practice, the iodine molecules will be touching one another (right-hand diagram); they have been drawn apart for the sake of clarity. The molecules of iodine are held together by London forces (see **Topic 7**).



▲ **fig E** Large crystals of iodine.



▲ **fig F** Sucrose.

Other molecular solids include sulfur ($S_8$), white phosphorus ($P_4$), Buckminsterfullerene ($C_{60}$, **fig G**), dry ice (solid carbon dioxide, $CO_2$, **fig H**), sucrose ($C_{12}H_{22}O_{11}$, **fig F**) and solid alkanes (e.g. paraffin wax).



▲ **fig G** Buckminsterfullerene.



▲ **fig H** Dry ice.

## PHYSICAL PROPERTIES OF MOLECULAR SOLIDS

Molecular solids will, in general, have low melting and boiling temperatures. In order to melt a molecular solid, it is not necessary to break the covalent bonds within the molecule (the intramolecular bonds); it is only necessary to overcome the intermolecular forces of attraction.

Since intermolecular forces of attraction tend to be much weaker than covalent bonds, little energy is required to either break down the lattice structure of the solid and cause it to melt, or to separate the molecules and cause the liquid to boil and vaporise.

Intermolecular forces tend to increase with both an increase in the number of electrons per molecule and also with increasing length of molecule. This means that a macromolecular solid such as poly(ethene) will have a much higher melting temperature than its monomer, ethene.




ethene
melting point = –169°C

poly(ethene)
melting point typically 120 to 180°C

▲ **fig I** Ethene and (poly)ethene.

## CHECKPOINT

**1.** Magnesium oxide ($Mg^{2+}O^{2-}$) has the same structure as sodium chloride ($Na^{+}Cl^{-}$).

(a) Draw a diagram to show the arrangement of the ions in a crystal of magnesium oxide.

(b) Explain why the melting temperature of magnesium oxide is higher than that of sodium chloride.

**2.** Explain the following observations:

(a) Magnesium and magnesium fluoride both have giant lattice structures containing ions. Solid magnesium conducts electricity, but solid magnesium fluoride does not.

(b) Silicon and phosphorus are both described as covalent substances, but silicon has a much higher melting temperature than phosphorus.

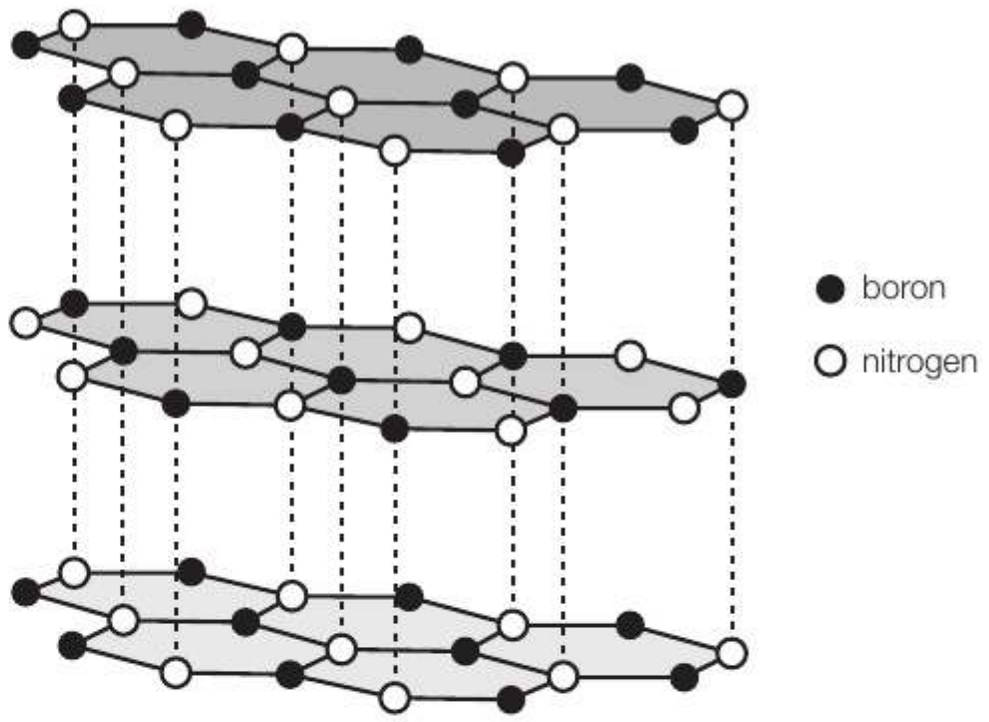
**3.** Hexagonal boron nitride has a structure similar to that of graphite.

Suggest why solid boron nitride can act as a lubricant in a vacuum, whereas graphite cannot.

**4.** Explain, in terms of their bonding and structure, why

(a) diamond is hard and graphite is soft, and

(b) diamond does not conduct electricity but graphite does.



# 3E 2 STRUCTURE AND PROPERTIES

SPECIFICATION REFERENCE 3.12

## LEARNING OBJECTIVES

- Predict the type of structure and bonding present in a substance from numerical data and/or other information.
- Predict the physical properties of a substance, including melting and boiling temperature, electrical conductivity and solubility in water, in terms of:
  - (i) the types of particle present (atoms, molecules, ions, electrons)
  - (ii) the structure of the substance
  - (iii) the type of bonding and the presence of intermolecular forces, where relevant.

## TYPES OF BONDING AND STRUCTURE

**Table A** shows the types of bonding and structure that exist in elements and compounds.

| BONDING | STRUCTURE | EXAMPLES |
|---|---|---|
| Metallic | giant lattice | Mg, Al, Cu, Zn |
| Ionic | giant lattice | NaCl, MgO, CsF |
| Covalent (including polar covalent) | giant lattice<br>molecular<br>macromolecular | C (diamond), C (graphite), Si, $SiO_2$, BN<br>$H_2O$ (ice), $I_2$, $P_4$, $S_8$, $C_{60}$, $C_{12}H_{22}O_{11}$ (sucrose)<br>polymers (e.g. poly(ethene)), proteins, DNA |

**table A** Types of bonding and structure.

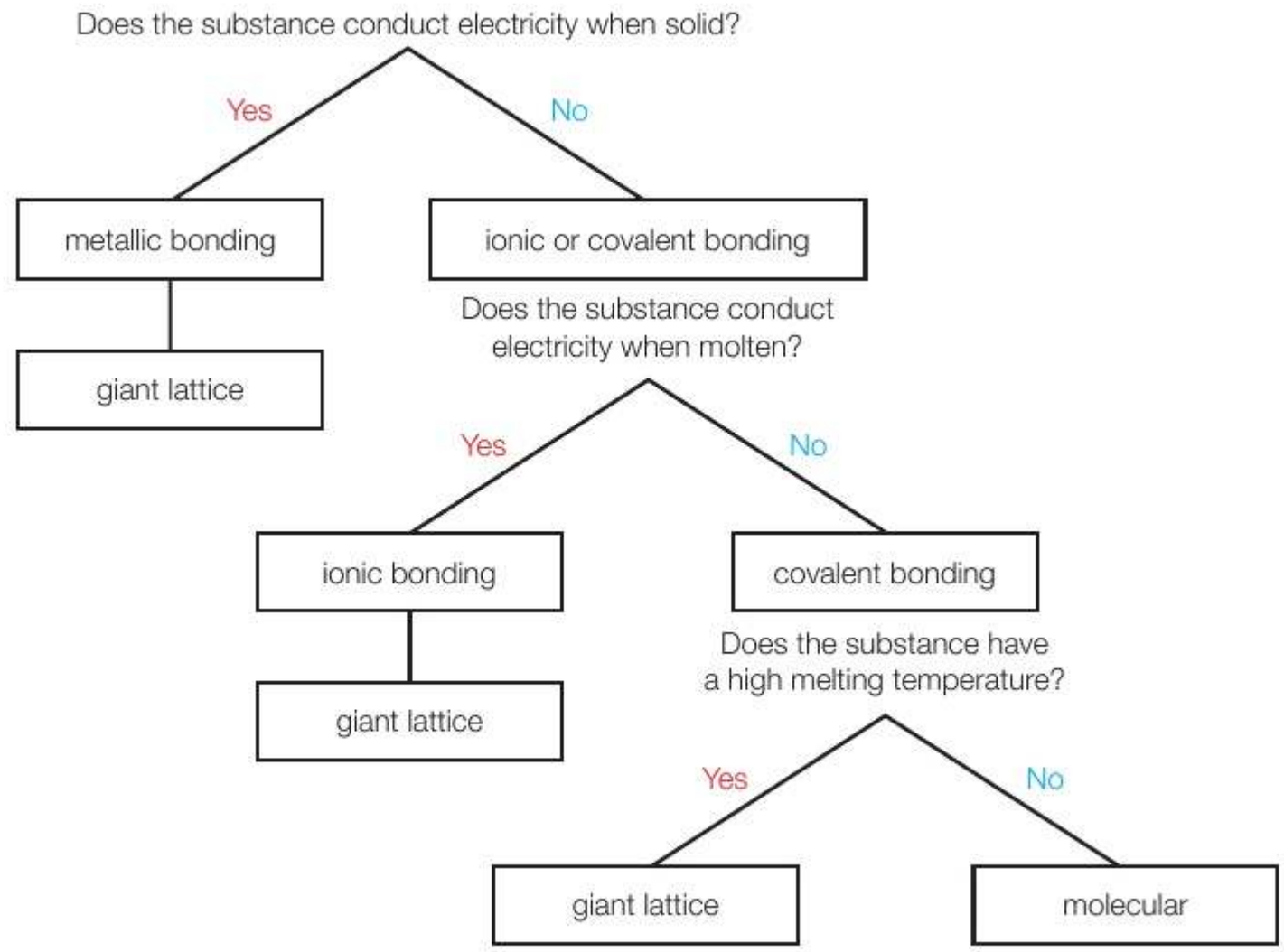
### EXAM HINT

If a question asks you to explain the properties of a substance, it is not enough simply to describe the bonding. You must refer to the particles involved and how they are arranged or behave. For example, metals conduct electricity because the delocalised electrons are free to move through the metallic crystal when a potential difference is applied.

## PREDICTING PHYSICAL PROPERTIES

The physical properties of a substance are determined by the type of bonding and structure it has.

**Fig A** allows you to determine the type of bonding and structure in a substance by considering some of its properties.




▲ **fig A** Flow chart for determining bonding and structure.

There will, of course, always be exceptions. For example, graphite has a giant covalent structure and yet it is a relatively good conductor of electricity when solid.

**Table B** gives a summary of the major properties of each type of structure.

| | GIANT METALLIC | GIANT IONIC | GIANT COVALENT | MOLECULAR |
|---|---|---|---|---|
| Particles present | positive ions and delocalised electrons | positive and negative ions | atoms | molecules |
| Type of bonding | metallic | ionic | covalent | covalent |
| Are there any intermolecular forces of attraction? | no | no | no | yes |
| Melting and boiling temperatures | fairly high to high | fairly high to high | high to very high | generally low |
| Electrical conductivity | good when solid and when molten | non-conductor when solid; good when molten | non-conductor | non-conductor |
| Solubility in water | insoluble unless the metal reacts with water, e.g. sodium | generally soluble, but with notable exceptions, e.g. AgCl, AgBr, AgI and $BaSO_4$ | insoluble | generally insoluble, but may dissolve if hydrogen bonding is possible (e.g. sucrose), or if the substance reacts with water (e.g. $Cl_2$)<br>(See **Topic 7** for an explanation of hydrogen bonding.) |

**table B** Structure and properties of a substance.

## CHECKPOINT

**1.** The table below gives some properties of four substances: A, B, C and D.

Analyse the data and then decide what type of bonding and structure is likely to be present in each substance.

| | MELTING TEMPERATURE / °C | ABILITY TO CONDUCT ELECTRICITY WHEN SOLID | ABILITY TO CONDUCT ELECTRICITY WHEN MOLTEN | SOLUBILITY IN WATER |
|---|---|---|---|---|
| **A** | 1083 | good | good | insoluble |
| **B** | 119 | poor | poor | insoluble |
| **C** | 2230 | poor | poor | insoluble |
| **D** | 801 | poor | good | soluble |

**2.** Each of the substances listed below has at least one property that is unusual for its type of bonding and structure. In each case, state the most likely bonding and structure, and identify the unusual property or properties. Suggest a possible identity for each substance.

(a) Substance **P**.

Melting temperature 813 °C; good conductor of electricity when solid and when molten; reacts with water to form ammonia gas and an alkaline solution.

(b) Substance **Q**.

Melting temperature 1414 °C; semi-conductor of electricity when solid; insoluble in water.

(c) Substance **R**.

Melting temperature 98 °C; good conductor of electricity when solid; reacts with water to form hydrogen gas and an alkaline solution.